Supreet Shukla

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.19
Kendriya Vidyalaya CRPF Mokamaghat, Patna	Class XII (Central Board of Secondary Education - CBSE)	2015	90.00%
Kendriya Vidyalaya CRPF Mokamaghat, Patna	Class X (Central Board of Secondary Education - CBSE)	2013	10.00%

M.TECH THESIS

Lattice-Based Signatures and Zero-Knowledge Proofs on Embedded Devices

- Optimized ML-DSA/Dilithium implementations for ARM Cortex-M4 microcontrollers, addressing the memory-latency trade-off in resource-constrained devices.
- Optimized the LaZer lattice-based ZKP library using matrix/vector streaming, compressed representations, shared workspaces, early rejection, and column-major computation to reduce peak memory usage.
- Profiled memory and execution overhead for lattice-based commitments, anonymous credentials, and Kyber ZK proofs, achieving up to 99.9% memory reduction for matrix-vector computation and 86.9% reduction in Kyber ZK proof peak memory.

PROJECTS

Post-Quantum Secure Multi-Client Communication System | *C, FreeRTOS, ARM Cortex-M4, POSIX Threads, Sockets, ML-KEM*

- Built a multi-client secure communication system in C with a concurrent server and multiple ARM Cortex-M4 clients running FreeRTOS.
- Implemented RTOS-based multitasking and inter-task communication using tasks, queues, synchronization primitives and interrupt-driven I/O.
- Integrated ML-KEM-based key establishment and authenticated encryption, and benchmarked communication latency, CPU utilization and memory footprint on Cortex-M4.

LLC Cache Replacement & Write Management | *ChampSim, C++, SPEC CPU*

- Implemented and evaluated LRU, SRRIP, DRRIP, and Random replacement policies at the LLC using SPEC CPU workloads.
- Developed probabilistic write-bypassing (10-30%) and set-remapping mechanisms to analyze cache utilization, write distribution, and memory traffic.

Real-Time Interrupt Management Framework | *C, ARM Cortex-M4, NVIC, UART*

- Designed and implemented a **real-time interrupt management framework** on STM32F446RE with prioritized, preemptive interrupt handling.
- Configured **hardware timers, GPIO, NVIC priorities, external interrupts and UART**, with three periodic tasks and a high-priority sensor-triggered task.
- Measured **ISR execution time, interrupt latency and jitter** using the ARM Cortex-M4 DWT cycle counter and validated interrupt behavior through GPIO waveform analysis.

EXPERIENCE

Research Assistant | Embedded Post-Quantum Cryptography | IIT Hyderabad

- Optimized ML-DSA/Dilithium and lattice-based ZKP implementations for resource-constrained ARM Cortex-M4

platforms using memory- and computation-efficient techniques.
- Benchmarked execution time and peak memory usage across lattice-based signatures, anonymous credentials, and Kyber ZK proofs.

Faculty | Computer Science & Engineering | iGate Bhilai

- Taught and mentored 700+ students over 1.5 years.
- Subjects: Engineering Mathematics, Database Management Systems, and Operating Systems; mentored GATE CSE students securing AIR 2, AIR 4, and AIR 6.

PUBLICATIONS

Geometry-Aware Knowledge Distillation for Peer-to-Peer Federated Learning (GAKD)

- Developed a representation-geometry-based knowledge distillation approach using supervised contrastive learning and relational knowledge distillation to mitigate representation drift and improve cross-client generalization.

Micro-ML-DSA: Optimized Post-Quantum Signatures for Resource-Constrained Devices

- Co-authored a paper proposing a memory-efficient ML-DSA implementation for resource-constrained Cortex-M4 devices, with optimized signing and improved computational efficiency.

TECHNICAL SKILLS

- **Programming:** C, C++, Python, Bash, SQL
- **Embedded Systems:** ARM Cortex-M4, Embedded C, Memory-Constrained Systems
- **RTOS & Concurrency:** FreeRTOS, Multithreading, Task Scheduling, Synchronization, Inter-Task Communication
- **Computer Architecture:** ARM, RISC-V, Cache Optimization, SIMD
- **Security & Networking:** Lattice-Based Zero-Knowledge Proofs, Lattice-Based Signatures (ML-DSA), Anonymous Credentials, OpenSSL, TLS/DTLS, X.509
- **Tools & Simulation:** Linux, Git, CMake, Make, GDB, Docker, ChampSim, Vim

RELEVANT COURSES

- **Systems:** Advanced Computer Architecture, Operating Systems, Embedded Systems, Hardware Architecture for Deep Learning
- **Data Science:** Fraud Analytics, Representation Learning
- **Cryptography & Security:** Cryptography, Advanced Cryptography, Network Security, Post-Quantum Cryptography, Hardware Security

POSITIONS OF RESPONSIBILITY

PG Placement Head | Office of Career Services, IIT Hyderabad

- Directing end-to-end placement and internship operations for **2,500+ students and 300+ organizations**, leading a **400-member team** and implementing **OCS Portal enhancements** to streamline workflows for students and coordinators.
- Led **Career Connect 2026** with **30+ companies** and **Japan Career Day 2026** with **20+ Japanese firms**, managing end-to-end event execution and industry engagement.

President | Student Council, Acharya Narendra Dev College, University of Delhi

- Led the Student Council, coordinating student initiatives, representing student interests, and collaborating with faculty and administration to execute campus-wide programs